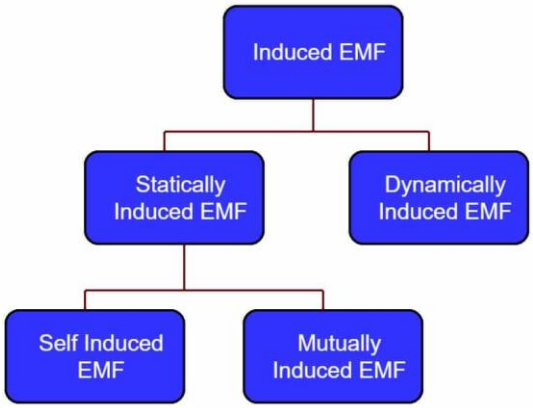
Chapter 48 - Electromagnetic Induction
Calculus Physics II

Electro-magnetic induction (EMI) – A voltage (emf or potential) is induced in a conductor moving in a magnetic field or a potential is induced in a coil of wire that is placed in a changing magnetic field. This induced emf is a result of the induced electric field generated by the changing magnetic field. The induced emf can produce a current in a circuit exactly the same as a battery or power supply.

- A changing Magnetic Field creates an Electric Field.

V_{emf} - is the ideal electric potential (voltage) produced by an electrical power source or is induced by Electromagnetic induction.

There are two main methods to produce an induced potential, Statically or dynamically.



Statically induced emf - is produced without any physical movement of the conductor or magnetic field, relying solely on changing magnetic flux linking the coil. An alternating current produces a changing magnetic flux, which links a stationary coil. An example is an electrical transformer.

- **Self-Induced:** EMF induced in a coil due to the change of flux produced by the same coil.
- **Mutually Induced:** EMF induced in one coil due to the changing flux produced by a neighboring coil.

Dynamically indued emf - is produced when there is relative motion between the conductor and the magnetic field. The conductor moves across the magnetic flux lines, which changes the magnetic flux in the device. An example is an AC generator.

Induced current - The current that is produced in a conductor when there is an induced emf (electric potential) present in that circuit.

Induced V_{emf} by a Change in the Magnetic Flux (Φ) - If a conducting loop, coil or ring experience a changing magnetic field inside the loop, by changing the area of the loop, the orientation relative to the field or changing the strength of the magnetic field, this will produce an electric field in the conducting wire. This electric field creates the electric potential and moves the charge around the wire loop if it is a complete circuit.

Magnetic Flux (Φ_M) - is the total magnetic field strength that passes through a defined area. The magnetic flux is the product of the area of a loop times the magnetic field through the inside of that loop. The orientation of the area to the magnetic field can affect the amount of the magnetic field getting through the area. Units of magnetic flux are Weber's (Wb).
1 Weber = 1 Tm^2

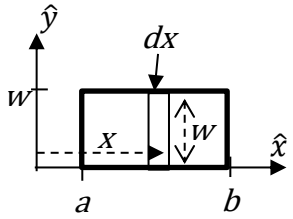
$$\Phi_M = \oint \vec{B} \bullet \hat{n} dA$$

Φ_M – Magnetic flux (Tm^2)
$\vec{B}$ – magnetic field strength (T)
$\hat{n}$ – Area unit vector and is perpendicular to the surface area (no units).
dA – Incremental surface area (m^2).

The angle (φ) is measured from a line that is drawn normal (perpendicular) to the surface area ($area\ unit\ vector\ \hat{n}$) relative to the magnetic field vector.

Example: For a magnetic field that is changing along the z-axis proportional to $\frac{1}{x}$ along the x axis. Determine the magnetic flux through the rectangular wire with a width "w" and length b – a. Knowing that $\hat{n}$ is in the $\hat{z}$ direction $\hat{z} \bullet \hat{n} = 1$

$$\vec{B} = \left(\frac{\mu_o I}{2\pi x}\right)\hat{z}$$



. Where dA = wdx

$$\Phi_M = \oint \vec{B} \bullet \hat{n} dA = \int_a^b \left(\frac{\mu_o I}{2\pi x}\right) w dx = \frac{\mu_o I w}{2\pi} \int_a^b \frac{1}{x} dx$$

$$\Phi_M = -\frac{\mu_o I w}{2\pi} [\ln(b) - \ln(a)] == -\frac{\mu_o I w}{2\pi} \left[\ln\left(\frac{b}{a}\right)\right]$$

For a uniform magnetic field - The magnetic flux through area A is defined to be the product of B⊥ and A where B⊥ is the component of $\vec{B}$ perpendicular to the surface area A.

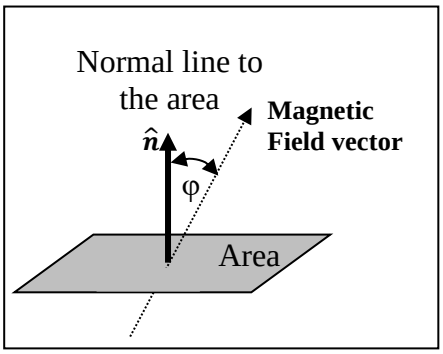
$$\Phi_M = \oint \vec{B} \bullet \hat{n} dA = \left|\vec{B}\right| \cos(\varphi) \int dA$$

$$\Phi_M = \vec{B} \bullet \hat{n} A = \left|\vec{B}\right| \cos(\varphi) A$$

φ - is the angle between the magnetic field vector and the normal (perpendicular) to the area ($\hat{n}$).

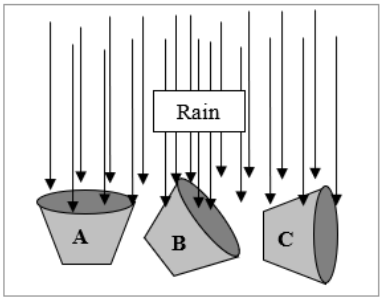
$|\vec{B}|$ - is the magnitude of the magnetic field

A – is the surface area made by conducting wire (inside)



- ❖ The larger the magnetic field strength the larger the flux.
- ❖ The larger the area the larger the flux.
- ❖ The maximum flux through an area is when the normal to the surface of the area is along the magnetic field lines.
- ❖ The minimum flux or zero flux is when the normal to the surface of the area is perpendicular to the magnetic field lines.

Example: For a bucket to collect the maximum amount of rain in a bucket, which bucket will collect the most rain or which bucket will have the most rain flux passing through the buckets opening?



Bucket A has the normal to its top surface area pointing along the line of the rain drops so **φ = 0°** and Acosφ = maximum, so the most rain will be collected using bucket A

(Rain flux **Φ = [Rain field]A)**

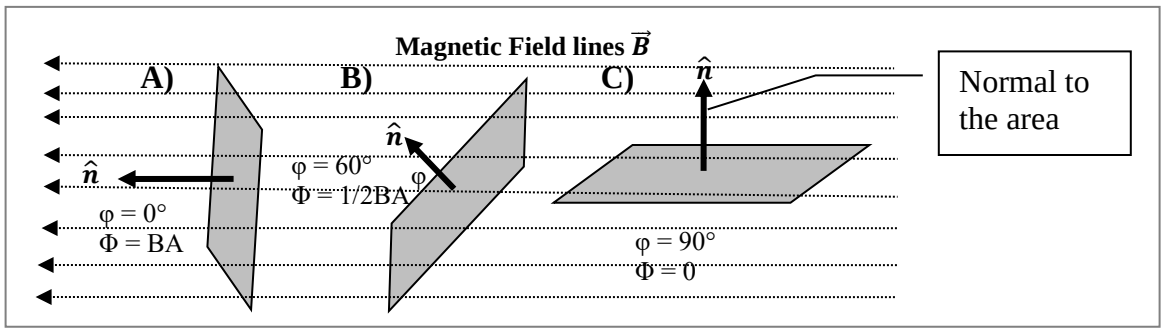
Bucket B has the normal at an angle to the rain so some rain gets in but not the maximum

$$\Phi_{Rain} = [(\text{Rain field})\cos\varphi]A$$

Bucket C has the normal to its top surface area pointing perpendicular to the line of rain drops so **φ = 90°** and Acosφ = 0, no rain will be collected **Φ = 0°**

Note: If it starts to rain harder (larger magnetic field) then the larger the flux through the area.

Example: Determine the magnetic flux through a rectangle with area of 0.2 m^2 and a magnetic field strength of 4 T.



A) The normal line to the area is pointing along the magnetic field line therefore the angle φ is 0° and the magnetic flux through the area is maximum (cos(0°) = 1).

$\Phi_M = BA\cos\varphi = (4\ T)(0.2\ m^2)(\cos(0°)) = 0.8$ Webers

B) The normal line to the area is pointing at an angle of φ = 60° to the magnetic field line therefore the magnetic flux through the area is reduced by half (cos(60°) = 0.5).

$\Phi_M = BA\cos\varphi = (4\ T)(0.2\ m^2)(\cos(40°)) = 0.40$ Webers

C) The normal line to the area is pointing at an angle of φ = 90° to the magnetic field line therefore the magnetic flux through the area is zero no magnetic field lines are going through the area of the wire (cos(90°) = 0 Webers

$\Phi_M = BA\cos\varphi = (4\ T)(0.2\ m^2)(\cos(90°)) = 0$ Webers

<u>Faraday's Law</u> – states that a changing magnetic flux create an induced electric field. This induced electric field induces a V_{emf} or voltage in a conductor, proportional to the rate of change of magnetic flux (Φ_M) and the number of turns (N) in a coil.

For a coil of wire with "N" number of loops (wraps or turns of wire) that encloses an area A, and a changing magnetic flux $\Delta\Phi_M$ through the coil, and the changing flux occurs during a time Δt then the average emf induced between the two ends of the coil is given by

$$V_{emf} = -\frac{d\Phi_M}{dt} = \frac{d}{dt}\left[\oint \vec{B} \bullet \hat{n} dA\right]$$

For a uniform magnetic field

$$V_{emf} = -N\left(\frac{\Delta\Phi_M}{\Delta t}\right)$$

V_{emf} - is the induced emf - electric potential (Volts).
N - number of loops of wire or turns of wire in the coil.
$\Delta\Phi_M$ = is the change in magnetic flux.
φ - is the angle between the direction of the magnetic field vector and the normal (perpendicular) to the area ($\hat{n}$).

- A potential is <u>only induced</u> if there is a <u>changing magnetic flux.</u>
- This equation applies to all methods of generating an emf!
- The minus sign indicates that the induced emf opposes the change in the flux which produces it, and is known as Lenz's Law.

For a changing magnetic flux a potential can be induced by changing the magnetic field strength, the area of the conductor or by changing the angle of the conductor relative to the magnetic field.

$$\Delta\Phi_M = \Phi_f - \Phi_i = B_f A_f \cos\varphi_f - B_i A_i \cos\varphi_i$$

- The induced emf will be larger for faster changes in the magnetic field.
- When the magnetic field strength, area of wire or angle of loop are not changing then there is no induced emf.
- Work is required to produce the changing field and the amount of work/energy put into creating this change is the amount of energy available in electrical energy.

Lenz's Law - The induced emf resulting from a changing external magnetic flux has a polarity that induces a current whose direction is such that the induced magnetic field from the induced current opposes the original external flux change.

Induced magnetic field – The magnetic field, which is produced by an induced current, always opposes the change in the external magnetic field.

The ***induced magnetic field*** is set up to opposes the change in the external magnetic field.

- ❖ For an increasing external magnetic field, the induced magnetic field is created in the opposite direction as the external magnetic field to reduce the magnitude of the external field and opposes the external field getting stronger.

- ❖ For a decreasing external magnetic field, the induced magnetic field is created in the same direction as the external magnetic field to increase the magnitude of the external field and opposes the external field getting weaker.

- ✓ The direction of the induced current is obtained by using the righthand rule # 2 and the direction the induced magnetic field must be to oppose the change in the external magnetic field!

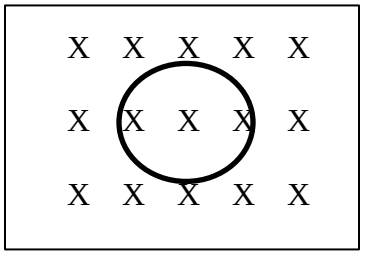
Determining the induced current direction - the induced emf produces a current that creates an induced magnetic field that always opposes the change in the external magnetic field!

1) Determine whether the magnetic flux that penetrates inside a coil is increasing or decreasing.
2) Determine direction of the external Magnetic Field lines.
3) Find what the direction of the induced magnetic field must be so that it can oppose the change in the external magnetic field.

Increasing External magnetic field → induced field opposite direction as external field
Decreasing External magnetic field → induced field same direction as external field

4) Having found the direction of the induced magnetic field using the righthand rule # 2 to determine the direction of the induced current that would produce a induced magnetic field in this direction.

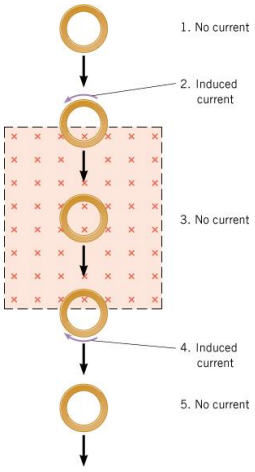
Example: A conducting ring is placed in an external magnetic field that is pointing into the page as shown.



A) The external magnetic field ($\vec{B}$) directed into the page, starts to increase in strength; the change in the external field is getting stronger inside the ring so the induced current is created to induce a magnetic field that makes the external field weaker (oppose the external field getting stronger), to do this the induced magnetic field ($\vec{B}_{\text{induced}}$) is in the direction that is opposite that of the external field, the induced field subtracts from the external magnetic field, the would make the induced magnetic field direction out of the page. Using RHR # 2, for a current to produce a magnetic field out of the page inside the ring the current must be in the counterclockwise direction.

B) If the external magnetic field inside the loop is still directed into the page but is decreasing in strength, the induced current would produce an induced magnetic field that adds to the external field (opposes external field getting weaker), which would be directed into the page. To produce this induced field the current direction would need to be in the clockwise direction.

Lenz's Law Example: Loop moving through a magnetic field directed into the page.



1. Inside the loop there is no magnetic field so there is no induced magnetic field
2. The loop goes from no magnetic field to some field inside the loop so there is a change in magnetic field, and it is getting larger. The induced magnetic field must be created to oppose the field getting larger so it is set up to subtract from the external field, therefore is opposite direction as the external field, the induced current would be counterclockwise, and this would induce a magnetic field that is out of the page.
3. The ring is moving through a portion of the field that is uniform (not changing) therefore there is no induced current or field because the magnitude of the external field through the ring is not changing
4. The loop goes from a section with a magnetic field to a section with no field so the magnetic field is changing, and it is getting smaller so the induced current produces a field that opposes the change in the external field so it must add the field to keep it from getting smaller. So the induced current would create a magnetic field in the same direction as the external field, therefore the current must be in the clockwise direction.
5. Same as # 1

Faraday's Law Example: The magnetic field is decaying over time in the $\hat{z}$ direction by

$$\vec{B} = \left(B_i e^{-t\lambda}\right)\hat{z}$$

Where B_i = 0.5 T is the initial magnitude of the magnetic field and λ is the decay constant λ = 0.2/s. The area of the coil A = 0.4 m^2 is not changing is oriented perpendicular to the magnetic field and has N = 100 loops. The magnetic field is not changing over the area and the area of the coil is not changing therefore $\hat{n}\bullet\hat{z}$ = 1

$$\oint \vec{B} \bullet \hat{n} dA = \left|\vec{B}\right| A$$

The induced potential is given as

$$V_{emf} = -N\frac{d\Phi_M}{dt} = \frac{d}{dt}\left[N\oint \vec{B} \bullet \hat{n} dA\right] = \frac{d}{dt}\left[N\left|\vec{B}\right|A\right] = \frac{d}{dt}\left[NB_i e^{-t\lambda} A\right] = \lambda\left[NB_i e^{-t\lambda} A\right]$$

Determine the induced potential at t = 5 s

$$V_{emf} = N\lambda A B_i e^{-t\lambda} = 1.47\ V$$

Example: A circular loop of wire is placed perpendicular to a constant magnetic field B, the change in the area (A) of the loop is given by $A = A_o \cos(2\pi f t)$, were b and f are constants, determine the equation for the induced V_{emf} in the loop.

The magnetic field is not changing with area and the area is not changing therefore

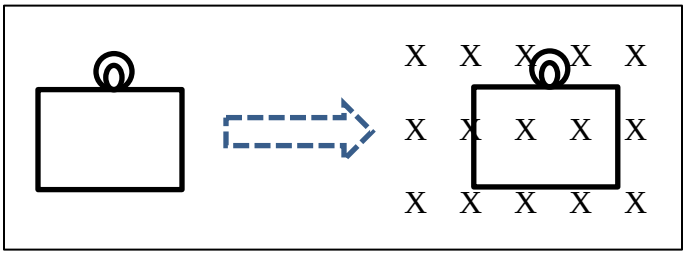
$$\oint \vec{B} \bullet \hat{n} dA = \left|\vec{B}\right| \int dA = \left|\vec{B}\right| A$$

The induced potential is given as

$$V_{emf} = -\frac{d\Phi_M}{dt} = \frac{d}{dt}\left[\oint \vec{B} \bullet \hat{n} dA\right] = \frac{d}{dt}\left[\left|\vec{B}\right|A\right] = \frac{d}{dt}\left[\left|\vec{B}\right|(A_o \cos(2\pi f t))\right]$$

$$V_{emf} = \left|\vec{B}\right| A_o (2\pi f)(\sin(2\pi f t))$$

Example of uniform field: A rectangular wire (A = 0.5 m^2 and N = 1 loop) wired in series with a light bulb (R = 5 Ω) is in space with no magnetic field, then is pulled into the magnetic field inside of 0.4 T in a time of 40 ms.



A) Determine the induced voltage in the ring

$\Delta\Phi_M = \Phi_f - \Phi_i = B_f A_f \cos\varphi_f - B_i A_i \cos\varphi_i$

$\Delta\Phi_M$ = 0.2 Wb – 0 Wb = 0.2 Wb

$$V_{emf} = -\frac{N\,\Delta\Phi_M}{\Delta t} = -\frac{(1)(0.2Wb)}{0.04s} = -5\,V$$

B) Determine the current in the bulb as the rectangle is being moved?

V = IR therefore I = V/R = 1 A

C) Which direction is the current in the rectangular loop? The external magnetic field is into the page and getting stronger (from zero to 0.4 T), the induced field created by the induced current must opposes this change therefore the induced field will be in the opposite direction to the external field which is out of the page. In order for a current in the loop to produce a field out of the page it must be in the counterclockwise direction.

D) Once the rectangular loop is in the field and not moving what is the induced current in the lamp? It is zero because there is no change in magnetic flux

E) If the rectangular loop that is in the magnetic field is pulled out of the field in 40 ms what would be the induced current and in what direction?

$\Delta\Phi_M$ = 0 Wb - 0.2 Wb = - 0.2 Wb

$$V_{emf} = -\frac{N\,\Delta\Phi_M}{\Delta t} = -\frac{(1)(-0.2Wb)}{0.04s} = 5\,V \qquad I = V/R = 1\,A$$

The current would be the same because it will experience the same change in flux over the same amount of time therefore I = 1 A

The direction of the current would be opposite or clockwise because the loop would now want to oppose the change in the external field which is getting weaker therefore the induced field would add to the external field.

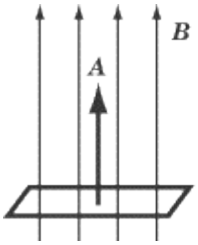
Example: A rectangular loop of wire is placed in a magnetic field with the normal to the area directed along the magnetic field lines.

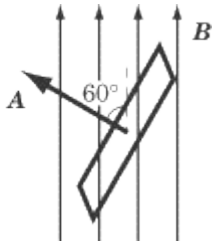
A) What is the induced voltage in the loop (A = 0.1 m^2 and N = 10 loop) of wire that starts with its normal parallel to the magnetic field (B = 0.4 T) and is rotated by 60º in 0.02 s?

$\Delta\Phi_M = \Phi_f - \Phi_i = B_f A_f \cos\varphi_f - B_i A_i \cos\varphi_i$

$\Delta\Phi_M = 0.02\ Wb - 0.04\ Wb = -0.02\ Wb$

$$V_{emf} = -\frac{N\,\Delta\Phi_M}{\Delta t} = -\frac{(10)(-0.02Wb)}{0.02s} = 10\ V$$





B) What is the induced voltage in the loop (A = 0.1 m^2 and N = 10 loop) of wire that starts with its normal at 0° to the magnetic field (B = 0.4 T) and is rotated to 180º in 0.02 s?

$\Delta\Phi_M = \Phi_f - \Phi_i = B_f A_f \cos\varphi_f - B_i A_i \cos\varphi_i$

$\Delta\Phi_M = (-0.04\ Wb) - (+0.04\ Wb) = -0.08\ Wb$

$$V_{emf} = -\frac{N\,\Delta\Phi_M}{\Delta t} = -\frac{(10)(-0.08Wb)}{0.02s} = 40\ V$$

<u>Motional emf</u> - When an electrically conductive material moves through a magnetic field (but not along a field line) the charged particles that are free to move in that material feel a force because of their charge due to the fact that they are moving in a magnetic field. The positive charged particles are forced to move in one direction in the conducting material and the negative charged particles are forced to move in the opposite direction (uses Right hand rule # 1 for direction of force on the positive charge). This charge separation or emf is due to the motion of the conductor in a magnetic field.

- An emf is induced in a conductor moving in a magnetic field, this produces the same results as a batteries chemical charge separation, a circuit would not know the difference.

  - The charge separation will be maximum when the electrostatic force between the separated charges is equal to the induced magnetic force separating them.
  - When the conductor stops moving, the force keeping the charges separated is gone and the charge will flow back together and become neutral again.
  - Work is required to produce this charge separation and the amount of work/energy put into creating this charge separation is the exact amount of energy available in electrical energy or electric potential energy per unit charge for the circuit.

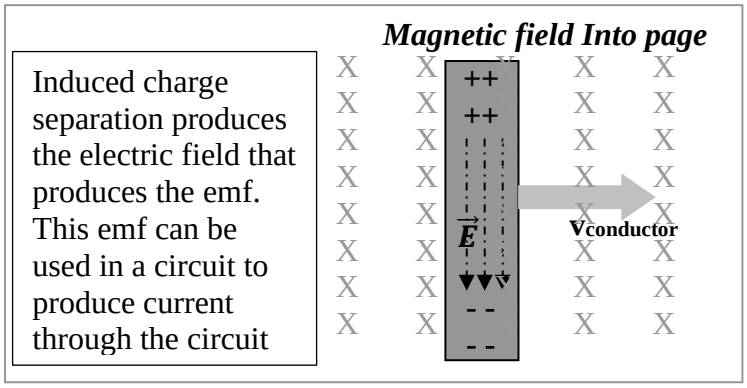
<u>V_{emf} induced by a conductor moving in a magnetic field</u>- Motional emf - When a straight conductor of length (L) is moving across a magnetic field strength ($\vec{B}$) with a velocity ($\vec{v}$) the induced emf in the conductor is given by

$$\mathbf{V_{emf}} = |\vec{v}|\left|\vec{B}\right|\mathbf{L}\sin\boldsymbol{\varphi}$$

V_{emf} - the electric potential (or emf) induced across the conductor (Volts).

$\vec{B}$ - magnetic field strength (Tesla).
$\vec{v}$ - magnitude of the velocity of the conductor (m/s).
L - length of the conductor in the magnetic field (m) perpendicular to the motion.
φ - angle between the velocity vector and the magnetic field direction (degrees).

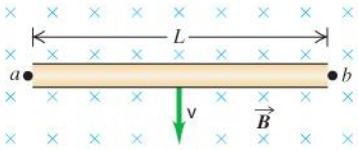
- The larger the $\vec{B}$ field the larger the induced emf.
- The faster the conductor is moving the large the induced emf.
- The more of the conduct that is in the $\vec{B}$ field the larger the emf.
- For maximum induced emf the conductor must be moving perpendicular to the magnetic field (φ = 90°) then V_{emf} = vBL.
- There is NO induced emf if the conductor is moving along the magnetic field lines.



Example: A car has a conduction metal axel that is L = 2 m long, and it is moving with a velocity of $|\vec{v}|$ = 27 m/s through the earth's magnetic field ($|\vec{B}|$ = 3 x 10^{-4} T) at an angle of φ = 60° between the velocity and the magnetic field lines, what is the induced potential across the axel due to the motion of the car through the field?

V_{emf} = vBLsinφ

V_{emf} = (27 m/s)(3 x 10^{-4} T)(2 m)sin(60°) = 0.014 V



If the car's axel was moving perpendicularly to the field what would the induced potential be?

V_{emf} = vBLsinφ = (27 m/s)(3 x 10^{-4} T)(2 m)sin(90) = 0.016 V

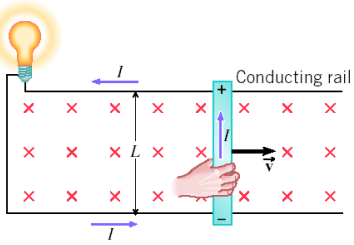
Motional emf and Electrical Energy (Conservation of Energy) - In order to move a conduction rod in a magnetic field there must be a force applied over a distance, this is the work done by the applied force. If there is a current induced in the rod, then this induced current is in a magnetic field which produces a magnetic force to oppose the motion.

- The work done by the applied force is equal to the electrical energy supplied to the circuit.

Example of lighting a light bulb using motional induced emf - A conducting bar is pushed ($\vec{F}_{applied}$) along a conducing loop to the right in an external magnetic field at a constant velocity the charge in the conductor will move creating a current due to the induced potential. This induced potential will produce a current that will light the light bulb. This induced current is moving through an external magnetic field, because of this it will produce a force ($|\vec{F}_M|$= ILBsinφ) due to the current flowing in the wire in a magnetic field. This induced force ($\vec{F}_M$) will oppose the applied force ($\vec{F}_{applied}$) that is moving the conductor down the wire. The faster the conductor is pushed the larger the current produced and the larger the force the magnetic field produces to oppose that motion.

The applied force that moves the bar is opposed by a force that is created from the current flowing through the bar.

$$\vec{F}_{applied} = -\vec{F}_M$$



- This is not a frictional force or other resistive forces; it is a force that is a result of the induced current flowing in a wire in a magnetic field.

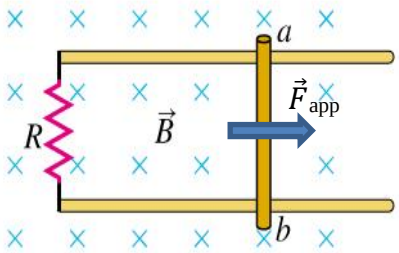
If there were no magnetic force acting on the conductor and if friction is ignored then there would be no force required to move the conductor along the wire, the energy created by the induced potential would take no work to create, it would not take any energy to create energy, this would violate the conservation of energy!

- ✓ Energy is conserved because the work done by the force pushing the conductor is equal but opposite to the work done by the force of the induced current as the conductor is pushed!

Example: Two conducting rails with a resistor (R = 5 Ω) connecting them have a conducting rod placed cross the rails forming a circuit, the rails are 0.4 m apart (L = 0.4 m). The rod is moved by an applied force ($\vec{F}_{app}$) directed to the right and perpendicular (φ = 90°) to a magnetic field directed into the page. If the rod moves at a constant velocity of 3 m/s and the field strength is 2 T.

A) Which way will the charge separate relative to points a and b?

Using the right-hand rule # 1 the motion of the charge is to the right; the magnetic field is into the page therefore the force on the positive charge will be up towards point "a" and the negative will be down towards point b.



B) What is the induced emf across the rod?

$V_{emf} = vBL\sin\varphi = (3\ m/s)(2\ T)(0.4\ m)(1) = 2.4\ V$

C) What is the current following through the resistor during this motion?

$V = IR$ therefore $I = V/R = 2.4\ V/5\ \Omega = 0.48\ A$

D) Because there is a current flowing through the rod in a magnetic field it will experience a magnetic force, what is the magnitude of the magnetic force due to the current and in what direction?

$F_m = ILB\sin\varphi = (0.48A)(0.4\ m)(T)(1) = 0.384\ N$

If the current is flowing from point b to point a then using the right-hand rule # 1 and pointing in the direction of current flow, with the magnetic field into to the page the magnetic force is to the left.

E) How much power is dissipated by the resistor in the circuit?

$P = IV = (0.48\ A)(2.4V) = 1.152\ W$

F) How much energy is used by the resistor in 2 seconds?

$P = Energy/time$ therefore $Energy = Pt = 2.304\ J$

G) How much work does the applied force do in moving the rod at 3 m/s for 2 seconds? (ignore frictional forces)

velocity $v = \Delta x/\Delta t$ therefore
$\Delta x = vt = (3\ m/s)(2\ s) = 6\ m$ there rod moves 6 meters in 2 seconds

$Work = F\Delta x = (0.384\ N)(6\ m) = 2.304\ J$

- This is exactly the same energy used by the resistor!

H) If the resistor was removed thus making an open circuit, what would happen to the magnetic force and the work done by the applied force? Once the circuit is open, current stops flowing and there is no longer a magnetic force due to the current in the rod. Because there is no magnetic force there will be no work required to move the rod.